FGSM — Goodfellow, Shlens, Szegedy (2015): 1-page summary

Adversarial examples — inputs perturbed imperceptibly to a human but classified incorrectly by a neural network — were first demonstrated by Szegedy et al. (2013). The prior explanation attributed them to extreme nonlinearity and overfitting: NNs were thought to have small, weird pockets in input space where predictions flip. Goodfellow et al. argue the opposite. Adversarial examples exist because neural networks behave **too linearly** in high-dimensional input spaces, not too nonlinearly. From this insight they derive the Fast Gradient Sign Method (FGSM), a cheap one-step attack. They also observe that adversarial examples often **transfer** across models — a perturbation crafted for one network frequently fools others, even with different architectures.

The linearity argument. Consider a linear model with output $w^T x$. Perturb the input as $\tilde{x} = x + \eta$, giving $w^T\tilde{x} = w^T x + w^T\eta$. Under an $L\infty$ constraint $||\eta||\infty \leq \varepsilon$ (each pixel changes by at most ε), the perturbation that maximizes $w^T\eta$ is $\eta = \varepsilon \cdot \mathrm{sign}(w)$. The activation shift is then $\varepsilon \cdot ||w||_1$, which grows linearly with input dimension n. So even though the per-pixel change stays bounded at ε and is imperceptible, the resulting shift in the model's pre-activation can be made arbitrarily large by increasing dimensionality. This is the core claim: linear behavior in high dimensions is *sufficient* to explain adversarial vulnerability — no nonlinear pathology required. Supporting this, the paper shows that even **logistic regression** is vulnerable to adversarial examples, which a nonlinearity-based explanation cannot account for.

Linear approximation for nonlinear models. Neural networks are nonlinear globally but approximately linear around any given input (first-order Taylor expansion of the loss). Replacing w with the loss gradient $\nabla_x J(\theta, x, y)$ and applying the same $L\infty$-optimal perturbation gives:

$x\_adv = x + \varepsilon \cdot \mathrm{sign}(\nabla_x J(\theta, x, y))$

Intuitively: ask which direction in input space most increases the loss, then take a single bounded step in that direction. One forward + one backward pass — far cheaper than iterative attacks — which is why FGSM remains the standard baseline.

Adversarial training as a defense. The paper proposes augmenting training with adversarial examples generated on-the-fly via FGSM, mixing the clean and adversarial loss:

$\tilde{J}(\theta, x, y) = \alpha \cdot J(\theta, x, y) + (1 - \alpha) \cdot J(\theta, x\_adv, y)$

This improves robustness on MNIST and acts as an effective regularizer. Adversarial training has since become the foundation for nearly all subsequent robustness work, including Madry et al.'s PGD-based training.

Takeaways. (1) Adversarial examples are a consequence of high-dimensional linearity, not nonlinear quirks. (2) FGSM is the $L\infty$-optimal single-step attack under a first-order linear approximation of the loss. (3) Transferability across models supports the linearity view: different networks learn similar linear directions. (4) Adversarial training is the paper's proposed mitigation and remains the default starting point for defense.